BINH GIA NGUYEN

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EDUCATION

FPT University *2022 – 2026*
B.Sc. in Artificial Intelligence Hanoi, Vietnam
GPA: 8.59/10 Graduated with Honors (Full scholarship)
Thesis: *LiteVLA-MS: Resolving Motion Under-Specification in Lightweight Vision-Language-Action Models* (9.5/10)

EXPERIENCES

Center for AI Research (CAIR), VinUniversity · Research resident *Apr 2025 – Present*
Upwork · AI Freelancer · Remote *May 2024 – Apr 2025*
BSM Labs · AI Engineer · Hanoi, Vietnam *Jun 2024 – Feb 2025*
Aimesoft · AI Intern · Hanoi, Vietnam *Mar 2024 – Jun 2024*

RESEARCH INTERESTS

I am interested in **Efficient AI** - specifically model compression, model pruning, efficient training and inference acceleration, and scalable deployment of large-scale models.

RESEARCH & PUBLICATIONS

CKA-Guided Layer Pruning for VLA Models 2025 – Present
Supervisor: Duy H. M. Nguyen VinUniversity / VinRobotics, Hanoi

> *Finetuning Vision-Language-Action Models Requires Fewer Layers Than You Think*
> **Binh G. Nguyen***, Trong-Bao Ho, Thien-Loc Ha, Khoa Vo, Philip Lund Møller, Quang Tan Nguyen, Long Dinh, Tuan Quan Dam, Vu N. Duong, Tung Minh Luu, Trung Le, Tran Nguyen Le, Minh Nhat Vu, An Thai Le, Ngan Le, Daniel Sonntag, James Zou, Jan Peters, Duy Minh Nguyen Ho, Vien Anh Ngo
> **RSS Workshop 2026** · clpvla.github.io

- Found that current state-of-the-art VLA models exhibit significant layer redundancy, which can be removed without significant performance drop
- Proposed CKA-Guided Layer Pruning (CLP), a training-free structural compression method for VLA models using Centered Kernel Alignment (CKA) layer similarity scores.
- Applied CLP to π_0, GR00T-N1.5, and SmolVLA; evaluated on LIBERO, RoboCasa, and SimplerEnv benchmarks and 10 real-world tasks with significant training time reduction, inference speedup and minimal performance drop.

Future-Oriented Conditioning for VLA Adaptation 2025 – 2026
Supervisor: Duy H. M. Nguyen VinUniversity / VinRobotics, Hanoi

> *FOCA: Future-Oriented Conditioning for Data-Efficient Vision-Language-Action Adaptation*
> Duc M. Nguyen*, Nghiem T. Diep*, **Binh G. Nguyen***, Trong-Bao Ho, Doanh Le, Tan Nguyen, Thien-Loc Ha, Tran Nhiem, Bao Thach, Nhat Tran, Tuan Anh Tran, Artur Habuda, Philip Lund Moeller, Tran Nguyen Le, Daniel Sonntag, Mathias Niepert, Khoa Doan, Vu Duong, Hung Ngo, Minh Vu, Duy MH Nguyen, An Thai Le, Vien Ngo
> **ICML 2026** · focavla.github.io

- Co-developed FOCA, a data-efficient adaptation method for VLA models via future-oriented action conditioning (co-first author).

- Finetune VLAs model include: GR00T N1.5, π_0, SmolVLA, SpartialVLA, DreamVLA, etc. as baselines to compare and World model Cosmos Predict 2.5 for video generations
- Regenerate large scale simulation dataset and extract segmentation mask and depth information include ROBOCASA, LIBERO.

Collaborative Research in VLA Ecosystem 2025 – Present
VinUniversity / VinRobotics, Hanoi

> *Start Right, Arrive Right: Asynchronous Execution via Initial Noise Selection*
> Trong-Bao Ho*, Quang Tan Nguyen*, Thien-Loc Ha*, **Binh G. Nguyen**, Viet-Thanh Nguyen, Long Dinh, Minh Nhat Vu, Duy Minh Ho Nguyen, An Thai Le, Vien Anh Ngo
> **CoRL 2026** · `paint-action-chunking.github.io`

> *EquiVLA: A General Framework for Rotationally Equivariant Vision-Language-Action Models*
> Thien-Loc Ha*, Quang Tan Nguyen*, Trong-Bao Ho*, Long Dinh, Minh Duc Nguyen, **Binh G. Nguyen**, Pham Tri Quang, Minh Nhat Vu, Duy Minh Ho Nguyen, An Thai Le, Vien Anh Ngo
> **RSS Workshop 2026** · `equivla.github.io`

> *Progress-Guided Preference Generation for Aligning Generative Control Policies*
> Tung Minh Luu*, Tri Ton, Minh Duc Nguyen, Thanh Xuan Nguyen, **Nguyen Gia Binh**, Quang Tan Nguyen, Vu N. Duong, Daniel Sonntag, Duy Minh Ho Nguyen, Vien Anh Ngo, Chang D. Yoo
> **CoRL 2026**

OPEN-SOURCE CONTRIBUTIONS

Tavish9/any4lerobot (1.2k stars) *2025*
A curated toolbox of data-conversion utilities for the LeRobot framework.
Contributed the RoboCasa → LeRobot dataset conversion pipeline.

HONORS & AWARDS

Champion — Funix GenAI Hackathon (Top 1 / 22 teams) *2024*
MindGPT: AI-powered mindmap generation web application.

100% Full Tuition Scholarship *2022 – 2026*
FPT University — Highest Math entrance score (86/90).

Honorable Student Award *2022 – 2026*
FPT University — Awarded every semester throughout undergraduate study.

SKILLS

Programming	Python, C++
Frameworks	PyTorch, HuggingFace Transformers, LeRobot, LangChain, Triton, ROS
Tools	Git, Docker, Weights & Biases, Linux, LaTeX
Research Interest	Efficent AI, Embodiment AI

REFERENCES

Dr. Duy H. M. Nguyen
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Principal Investigator, CAIR, VinUniversity
nhmduy.hcmus@gmail.com

Prof. Vu N. Duong
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Chief Scientific Officer, CAIR
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